

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division
FROM: Terri Shattuck, Environmental Quality Analyst, Water Resources Division
DATE: March 2, 2023
SUBJECT: Categorical Determination, Cadillac Castings

Per your request on December 5, 2022, with process and treatment information; along with additional process and production information provided on January 27, 2023, I have conducted a categorical determination for Cadillac Casting, Inc., which discharges process wastewater to the Cadillac Wastewater Treatment Plant (Cadillac WWTP).

Your review during a pretreatment inspection found that Cadillac Casting could be a metal caster and was not regulated under Title 40 of the Code of Federal Regulations (40 CFR), Part 464, the Metal Molding and Casting Effluent Limit Guidelines, and had asked for a formal categorical determination. This evaluation is based on information you provided via e-mails on December 5, 2022, January 27, 2023, and February 24, 2023. Information included a process description, process flow diagram, description of the on-site treatment systems, and e-mails. SDSs were not provided as no chemicals, including dyes, were used. The products produced by Cadillac Casting are ductile iron castings for the railroad industry. The plant was originally constructed in 1922, but the scrubbers were not installed until the mid-1990s.

The information provided describes two types of scrubber processes that discharge to the Cadillac sanitary sewer via two separate treatment systems. The two scrubber processes are the Melt Furnace Scrubber Operations and the Dust Collection Scrubber Operations. The discharge from the scrubbers ultimately combines with non-contact cooling water and sanitary discharge. There is a sampling point once all flows are combined. There is a slag quench process that mostly evaporates but overflow may occur. Any overflow goes into the Melt-side wastewater treatment system.

The Dust Collection Scrubber Operations are two stand-alone units that consist of sand and bentonite clay to catch the dusty air. The Dust-side filter press effluent goes into the sewer drain after going through treatment. The Melt Furnace Scrubber Operations is a three-stage scrubber system consisting of primary quench scrubber, venturi scrubber, and a demister stage. Each stage's discharge combines prior to entering the treatment system. The Melt-side filter press effluent goes into the sewer drain after going through treatment.

Based on the information provided, the processes used by Cadillac Casting are metal molding and casting operations of ductile iron castings. To make this determination, I consulted the October 1998, EPA Office of Compliance Sector Notebook Project: Profile of the Metal Casting Industry guidance document. Any process water from these operations and discharged to

sanitary is to be regulated under 40 CFR, Part 464.36, Subpart C – Ferrous Casting Subcategory for new pretreatment sources. Pretreatment standards for new sources are applicable since the scrubbers were installed after these rules were promulgated in 1985. The Dust Collection Scrubber falls under Part 464.36(c)(1), the Melting Furnace Scrubber under Part 464.36(f), and the Slag Quench under Part 464. 464.36(h). Part 464.31(a) – Specialized Definitions for Total Toxic Organics has additional pollutants required for the process used. The Dust Collection Scrubber must sample additional pollutants listed in Part 464.31(a)(2). The Melt Furnace Scrubber must sample additional pollutants listed in Part 464.11(a)(4). And the Slag Quench must sample additional pollutants listed in Part 464.11(a)(6).

Please contact me at 269-312-0548 or shattuckt@michigan.gov if you have any questions or comments regarding this categorical determination.

Attachments: December 5, 2022, E-mail and attachments
January 27, 2023, E-mail and attachments
February 24, 2023, E-mail and attachments

cc: Pamela Willard, WRD
File, Cadillac WWTP File,
Metal Molding and Casting Category